Fiscal Year:	FY 2017	**Task Last Updated:**	FY 07/05/2018
PI Name:	Ferri, James Ph.D.		
Project Title:	Particle Stabilized Emulsions and Foams		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	COMPLEX FLUIDS/SOFT MATTER--Complex Fluids		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	23219	**Congressional District:**	4
Comments:	NOTE: Previous affiliation was Lafayette College, Easton, PA until early 2018 when PI moved to Virginia Commonwealth University		
Project Type:	Ground	**Solicitation / Funding Source:**	2009 ESA AO Biology on Sounding Rockets and Physical Sciences on All Platforms
Start Date:	03/26/2015	**End Date:**	08/15/2017
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA GRC
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NNX15AE98G ; NNX10AV26G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	ED. NOTE: Continues research from grant NNX10AV26G, with same title. Overall aim of the PASTA (Particle Stabilized Emulsions and Foams) project is to develop a mechanistic description of the stabilization or destabilization of foams and emulsions, which can be accomplished by various combinations of surfactants, polymers, and particles. In this respect, we focus in particular on surface modification of (nano)particles by macromolecules (polymers) and biomacromolecules (proteins) and the concomitant changes to particle performance at the interface. Experimental work is contemplated on the three system levels -- the interface, the liquid film, and the formulated dispersion. Each level is interrogated so that structure-property relationships can be established along the hierarchy of increasing complexity. The comparison of foam and emulsion systems enabled through collaboration with the FOAM-C instrument, and the systematic consideration of surface modified (nano)particles as stabilizers/destabilizers will provide a new level of generalization. Experiments will be performed for the following hierarchies: 1. characterization of single interfaces with (nano)particles modified by polymers and proteins; 2. characterization of corresponding Gibbs (thin) and non-Gibbs (thick) emulsion films; 3. correlation between physico-chemical properties of (nano)particle modified interfaces and films, and; 4. correlation of (nano)particle physical chemistry, surface modification, interface and film properties with the behavior of disperse liquid-liquid systems (emulsions). In addition to the experimental program described above, theory will be developed which describes the behavior above in terms of the molecular and intrinsic constants associated with the materials of which they are composed. The composition of the consortium of participating teams (USA, Germany, Italy, Greece) is such that all aspects of the formulated problem are covered. The teams demonstrate a complementary expertise in form of the scientific fields and available experimental tools, as well as their capacity of tackling the theoretical side of the problems. Key experiments in microgravity conditions support the specific aims of this project. The basic processes involved in emulsions and foams, such as destabilization, mainly droplet aggregation and coalescence (destabilization of the film between droplets/bubbles), will be studied in the absence of segregation for model emulsions. In fact, during emulsion and foam destabilization, droplets larger than ten microns are rapidly segregating, making impossible on Earth the investigation of the above processes in controlled conditions. Microgravity experiments will then provide information about the features and characteristic times of these processes to be utilized to parameterize and validate the respective models. Since its inception, the PASTA project has sought to leverage the existing facilities on International Space Station (ISS) Fluid Science Laboratory (FSL). During the past three years, ESA (the European Space Agency) has refocused support in the Soft Matter Dynamics thrust to emphasize the FOAM-C facility which was originally developed to study coarsening and stability, respectively, of wet foams (multiple, polydisperse interfaces). As a result of shift in ESA priorities, in 2017 the PASTA team piloted the existing experimental module, FOAM-C, for suitability for emulsion dynamics studies.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	Early results using the FOAM-C test platform indicate the following: * The FOAM-C emulsification device is effective in producing emulsions, at least with large amounts of surfactant (concentrations above the cmc). Further tests are needed to access the effectiveness with surfactant concentrations below the CMC, which could be needed for some of the planned studies. It is noteworthy that emulsions were achieved with oils much more viscous than water showing the robustness of the design of the FOAM-C fluid-fluid dispersion generator, i.e., the piston. * The sensitivity of the diffusing wave spectroscopy (DWS) detector sufficient to measure an evolution of the emulsion. The data analysis must be however optimized to be effective for a quantitative analysis of the mechanistic phenomena. * Some challenges remain regarding the filling of the cell, with respect to the presence of air bubbles. * The cell material (polycarbonate) is not compatible with the long-term contact with alkanes (hexane, dodecane, etc.). Assuming the constraint of using the present foam cells, the studies should concern emulsions formulated using “milder” oils, which can be relevant for applications in food, pharmaceutical, or cosmetic industry. Collaborators in the PASTA team at AUTH-Greece have developed a substantially similar fluid-fluid dispersion generator to that of the FOAM-C apparatus. In summer 2018 with the new grant at Virginia Commonwealth University (VCU), the focus will rest on working on the AUTH prototype to measure the DWS signal for systems of interest that will be joining defined by the PASTA team. Further, technical discussions with collaborators in the PASTA network have identified the need to compare the DWS signal from the AUTH generator to the DWS data collected from systems produced in microfluidic emulsion generators in the VCU laboratories as a means to reduce (eliminate) polydispersity and characterize the experimental resolution of dynamic processes in liquid-liquid dispersions. Publications are in process. NOTE July 2018: Continued by "PASTA: Particle Stabilized Emulsions and Foams--80NSSC18K0453," due to Principal Investigator move to Virginia Commonwealth University in spring 2018. See that project for subsequent reporting.
Bibliography Type:	Description: (Last Updated: 12/03/2024)